Kevin Rajan

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Education

University of Illinois at Urbana-Champaign Urbana-Champaign, IL

Bachelor of Science in Engineering Physics, Minor in Computer Science *Aug. 2016 – May 2020*

- Certificate in Technology Commercialization
- Relevant Coursework: Applied Machine Learning (CS 498), Parallel Programming (CS 484), Operating Systems Design (CS 423), Compilers (CS 421), Computer Architecture (CS 233), Systems Programming (CS 241), Text Information Systems (CS 410), Computer Graphics (CS 418), Virtual Reality (CS 498)

Professional Experience

Founder & CEO Jun. 2024 – Present

zero, LLC *Remote*

- Architected end-to-end AI automation platform using Python, TypeScript, and REST APIs to enable LLMs to orchestrate complex workflows across 10+ enterprise applications, reducing manual intervention by 80%
- Built continuous research systems using Python, pandas, and scikit-learn to synthesize population-level health trends from 50K+ data points, informing preventative health solution development
- Designed and deployed reproducible Infrastructure as Code (IaC) environments using NixOS, Docker, and Kubernetes for on-premise GPU compute clusters, reducing operating costs by 50% ($15K/month savings)
- Prototyped digital twin simulations leveraging PyTorch and reinforcement learning to model AI factory workflows, achieving 95% accuracy in predicting system bottlenecks
- Led Agile product development lifecycle as sole technical founder, managing roadmap prioritization, sprint planning, and stakeholder communication while delivering MVP in 12 weeks

LLM Training Specialist Jan. 2025 – Present

Outlier (Scale AI) *Remote*

- Designed comprehensive evaluation rubrics for LLM assessment, authoring 500+ complex prompts in logic, mathematics, and physics to systematically identify reasoning failures and improve model performance by 15%
- Conducted red-teaming and adversarial testing across STEM domains, documenting 200+ edge cases and contributing to safety alignment improvements in production models (GPT-4, Claude)
- Performed large-scale preference labeling using RLHF, DPO, and RLAIF methodologies across 10K+ comparisons, achieving 94% inter-annotator agreement on quality rankings
- Developed multimodal evaluation frameworks integrating text, image, and tool-use tasks using Python and custom annotation pipelines, measuring grounding accuracy across 5 model architectures
- Built automated data validation scripts in Python to ensure annotation quality, reducing error rates by 30%

Ultralearning Sabbatical Jul. 2023 – Jun. 2024

Self-Directed *Naperville, IL*

- Engineered data-driven performance optimization system using Python, SQL, and Tableau to track 50+ daily metrics, applying ML-based correlation analysis to optimize cognitive and motor skill development
- Achieved top 1-5% ranking in competitive gaming environments (25M+ users) through systematic A/B testing of training protocols and iterative feedback loop optimization over 12 months
- Reverse-engineered complex game systems using statistical analysis to identify performance bottlenecks, documenting findings in 20+ technical reports

Senior Consultant – Technology Enablement Jun. 2021 – Jul. 2023

Slalom Consulting *Chicago, IL*

- Led DevOps modernization for Fortune 500 finance client ($4B+ revenue), deploying CI/CD pipelines using Jenkins, Terraform, and AWS across 12 teams, reducing deployment time by 70%
- Architected Infrastructure as Code solutions using Terraform and Ansible to provision AWS resources (EC2, S3, Lambda, RDS), managing 200+ cloud resources with 99.9% uptime SLA
- Designed automated AMI pipeline using Packer and Jenkins to publish biweekly golden Windows and RHEL images, ensuring CIS security compliance for 500+ developer workstations
- Championed DevOps training program for 500+ new hires using hands-on Agile methodology workshops, achieving 95% satisfaction scores and reducing onboarding time by 40%

- Orchestrated multi-cloud Kubernetes environments (AWS EKS, Azure AKS) for client demonstrations, containerizing 15+ microservices using Docker and Helm charts
- Served as Product Owner for preventative health platform initiative, leading Agile ceremonies (sprint planning, retrospectives) and managing product backlog with 100+ user stories in Jira
- Spearheaded centralized knowledge base creation using Confluence, curating 200+ technical documents and establishing foundation for AI-driven content generation

Software Development Engineer Intern (EC2 Fargate) Summer 2018
Amazon Web Services *Seattle, WA*

- Developed internal monitoring dashboard using Java and AWS CloudWatch to visualize container pool metrics, reducing mean-time-to-detection (MTTD) for customer issues by 40%
- Implemented automatic pool management system using Python and AWS Lambda to dynamically scale customer compute capacity, handling 10K+ container requests per hour during traffic spikes
- Contributed to EC2 Fargate service codebase using Java, participating in code reviews and deploying features to production serving millions of containers daily

IT Enterprise Architect Intern Jun. 2018 – Aug. 2018
BlueCross BlueShield (HCSC) *Chicago, IL*

- Developed data architecture recommendations using SQL and Python to support HCSC's enterprise data strategy, analyzing 10+ data sources and identifying integration opportunities
- Created executive presentations translating technical data insights into actionable business strategies, influencing $2M+ technology investment decisions
- Collaborated with cross-functional teams across 5 departments to document current-state architecture and propose modernization roadmap

iOS Engineer / UX Engineer Jul. 2017 – Feb. 2018
Prenosis Inc. *Champaign, IL*

- Designed and developed production iOS application using Swift and Firebase enabling physicians to manage patient lab results, vitals, and imaging for 1,000+ patients across multiple healthcare facilities
- Implemented proprietary Prenosis Number risk stratification algorithm, reducing physician decision time by 25% through automated patient prioritization
- Built RESTful API integrations to sync patient medical history, prescriptions, and clinical notes in real-time, processing 5K+ daily transactions with 99.5% uptime
- Conducted user research with 15+ physicians to iterate on UX design, improving task completion rate by 35%

Software Engineering Intern May 2016 – Jul. 2016
Wipro Consulting *Chennai, India*

- Engineered ETL pipeline using Python and SQL to migrate data from IBM Cognos to Tableau, processing 500K+ records and reducing report generation time by 60%
- Developed data source extractor tool to aggregate information from 10+ distributed servers, implementing query optimization that improved data retrieval speed by 45%

Freelance iOS Developer / UI Designer Jan. 2016 – Jul. 2016
ByteBros. *Champaign, IL*

- Built full-stack iOS rental marketplace application using Swift, Firebase, and Node.js with Venmo/PayPal payment integration, serving 500+ users in pilot launch
- Implemented OAuth authentication using Facebook SDK and Firebase Auth, achieving 99% login success rate
- Designed UI/UX mockups in Sketch following iOS Human Interface Guidelines, conducting 10+ user testing sessions to validate design decisions

Leadership Experience

iOS Developer / Open Source Coordinator / Sponsorship Lead Aug. 2015 – May 2018
HackIllinois *Champaign, IL*

- Developed iOS application using Swift serving 15,000+ users (hackers, mentors, recruiters) during 36-hour hackathon, featuring real-time event check-ins, hardware checkout, and push notifications
- Raised $300K+ in sponsorships by building relationships with 120+ corporations including Google, Microsoft, and Facebook, negotiating contracts and managing sponsor deliverables

- Managed open source initiative coordinating 40 mentors nationwide to guide 500+ hackers in contributing to open source projects, resulting in 50+ merged pull requests to major repositories
- Led cross-functional team of 8 staff members using Agile methodologies to plan and execute logistics for Midwest's largest collegiate hackathon

Admin Committee / Corporate Committee Aug. 2016 – May 2017
Association of Computing Machinery (ACM) *Champaign, IL*

- Managed ACM website and Active Directory infrastructure serving 2,000+ members using Linux, Apache, and MySQL
- Coordinated 20+ tech talks with Fortune 500 companies, increasing student attendance by 50% through targeted marketing
- Secured $10K+ in corporate sponsorships for hardware and infrastructure upgrades

Class Coordinator / iOS Developer Jan. 2016 – Jul. 2016
Codepath University Partnership *Champaign, IL*

- Established official Codepath partnership with UIUC, coordinating logistics for iOS bootcamp serving 50+ students per semester
- Developed 10+ iOS applications using Swift, CocoaPods, and REST APIs including movie discovery app using TMDB API
- Mentored 20+ students in iOS development fundamentals, with 80% completing capstone projects

Projects

RoutinePlus | *Swift, HomeKit, Firebase, Figma, Core ML* Jan. 2017 – Jun. 2017

- Designed and developed intelligent iOS alarm application using Swift and Core ML to analyze calendar patterns and optimize wake times, achieving 90% user satisfaction in beta testing
- Integrated HomeKit APIs to automate 10+ smart home devices (thermostats, lights, coffee makers) based on user routines and location
- Prototyped UI/UX in Figma following Material Design guidelines, conducting 15+ user interviews to validate feature priorities

Flight Simulator | *WebGL, Unity, C#, Oculus SDK, JavaScript* Oct. 2016 – Dec. 2016

- Built 3D flight simulator in WebGL using custom shader programming and procedural terrain generation with Diamond-Square algorithm, achieving 60 FPS on standard hardware
- Extended to Unity VR with Oculus Rift support, implementing quaternion-based camera controls and physics simulation
- Open-sourced project with 50+ GitHub stars and contributions from 5 developers

Nanotechnology Research | *Python, MATLAB, Spectroscopy, Laboratory Research* Oct. 2016 – Dec. 2016

- Synthesized biocompatible carbon nanoparticles for medical imaging applications, conducting 100+ experiments to optimize polymeric coatings
- Analyzed absorption spectra using Python and MATLAB, publishing findings on nanoparticle passivation techniques

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, Swift, SQL, Bash, Haskell, Go
Backend & APIs: Node.js, Express, FastAPI, Flask, REST APIs, GraphQL, Microservices Architecture
Cloud & DevOps: AWS (EC2, S3, Lambda, EKS, RDS, CloudWatch), Azure, GCP, Terraform, Ansible, Docker, Kubernetes, Jenkins, GitLab CI/CD, GitHub Actions, Helm, ArgoCD
AI/ML & LLMs: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, RLHF, DPO, RLAIF, LangChain, OpenAI API, Claude API, Hugging Face, MLflow, Weights & Biases
Data & Analytics: PostgreSQL, MySQL, MongoDB, Redis, Tableau, PowerBI, Apache Spark, Kafka
Frontend & Mobile: React, Vue.js, HTML/CSS, Swift, Xcode, Firebase, CocoaPods, UIKit
Project Management: Agile/Scrum, Jira, Confluence, Sprint Planning, Backlog Grooming, Stakeholder Management
Design & 3D: Figma, Sketch, Photoshop, Unity, Unreal Engine, Blender, WebGL